

What's Involved in Planning the Prom?



The Student Council is involved in helping plan prom. They have to find vendors, budget costs, and sell tickets. They need to break even on their costs, but they're not looking to make a profit.

1. The cost of renting a space for up to 400 people is a \$700 flat fee. The cost of catering a dinner for x attendees is given by $C(x) = 300 + 32x$. Write an equation for the total cost, $T(x)$, of putting on a prom if these are the only two expenses.

$$T(x) = R(x) + C(x)$$

$$T(x) = 700 + 300 + 32x$$

$$= 1000 + 32x$$

2. What is the domain of $T(x)$? How do you know?

Use most restrictive domain!

$0 \leq x \leq 400$ since we can't have negative people + even though catering is not restricted, the rental space is.

3. The 12 student council members are all required to attend the prom, but they don't have to pay for a ticket.

- a. Write an equation, $N(x)$, that gives the number of paying attendees if x students attend the prom.

$$N(x) = x - 12$$

- b. Is it possible for the number of attendees at prom to be 11? What about 12? 20? What is a reasonable domain for $N(x)$ in this situation?

11 is not possible bc the 12 student council members must attend.
12 and 20 are possible but not probable.
 $12 \leq x \leq 400$

4. Write the expression for $\frac{T(x)}{N(x)}$ and explain what this means in the context of this problem.

Price per ticket is a quotient of 2 other functions!

$\frac{1000 + 32x}{x - 12}$ represents the cost of attending prom for each non student-council member.

5. What is the domain of this new function? How does this answer compare to your answer in 2?

$\frac{T(x)}{N(x)}$ is more restrictive than $T(x)$ and $N(x)$ individually.

$x > 12$, even though 12 is included in the domain for $N(x)$, it wouldn't make sense here because we get an undefined cost since no one is available to pay.

6. The Student Council wants to sell tickets for \$40. How many total attendees do they think will come to the prom?

$$40 = \frac{1000 + 32x}{x - 12}$$

$$40(x - 12) = 1000 + 32x$$

$$40x - 480 = 1000 + 32x$$

$$8x = 1480$$

185 people

Math Medic
 $x = 185$

Lesson 1.7 – Building Functions

QuickNotes

Functions can be added, subtracted, multiplied or divided to create new functions.

To determine the domain of the combined function, use values that satisfy BOTH domains (most restricted).

When dividing functions, make sure the denominator function $\neq 0$

Check Your Understanding

- The number of customers at Pumpkin Pi Bakery depends on the price of the pie. Market research says that the number of customers can be estimated by $C(p) = 110 - 4.4p$, where p is the price of a pie.
 - If the bakery charges \$10 per pie, how many customers will there be? How much money (revenue) will Pumpkin Pi Bakery make at that price? (Assume each customer buys only one pie)

$$C(10) = 110 - 4.4(10) = 66 \text{ customers.}$$

$$66 \text{ cust.} \cdot 10 \$/\text{pie} = \$660$$

- Write an equation for the revenue Pumpkin Pi Bakery will make when they charge p dollars per pie. What is a reasonable domain for this function?

$$R(p) = p \cdot C(p) = p(110 - 4.4p) = 110p - 4.4p^2$$

$$p \geq 0$$

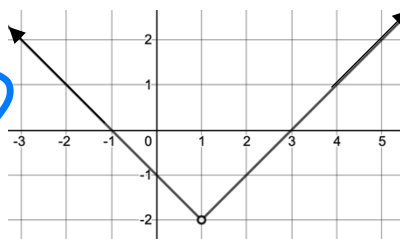
- The graph of $f(x)$ is shown below and a table of selected values is given for $g(x)$, a function defined for all real numbers.

- Let $h(x) = f(x)g(x)$. Find $h(0)$.

$$h(0) = f(0)g(0) = (-1)(-8) = 8$$

- What is the domain of $h(x)$?

$$\mathbb{R}, x \neq 1$$



x	$g(x)$
-3	0
0	-8
2	1
3	-4

- Find all x -values such that $h(x) = 0$.

$$\text{If } h(x) = 0 \text{ either } f(x) = 0 \text{ or } g(x) = 0$$

$$x = -1, 3, \text{ or } -3$$

- Let $f(x) = \sqrt{10 - x}$ and $g(x) = 8x + 7$.

$$D: x \leq 10$$

$$D: \mathbb{R}$$

- Find $f(x) - g(x)$. What is the domain of this function?

$$\sqrt{10 - x} - (8x + 7) \quad \text{Domain: } x \leq 10$$

- Find $\frac{f(x)}{g(x)}$. What is the domain of this function?

$$\frac{\sqrt{10 - x}}{8x + 7} \quad x \leq 10 \text{ AND } x \neq -7/8$$

$$(-\infty, -7/8) \cup (-7/8, 10]$$